

**Visual Search Experiment
PSY310 Lab in Psychology**

Lab Report

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GitHub Link :

Introduction

Visual search provides a controlled approach to examining how people tend to focus on important cues in the presence of distractions, which helps in the study of attention. It is a significant aspect in understanding attention and specifically, to efficiently identify relevant information and ignore irrelevant details. Visual search experiments are used to determine and investigate these processes by adding various variables, for example similar features, presence of a target, etc. The experiment uses proper frameworks to test how attention selects proper stimuli.

The way attention is used has a direct connection to the effectiveness of visual search; because visual search efficiency is a direct measure of how much attentional effort is needed to find a target. For example, a unique and obvious target would indicate a preattentive process and very less focused attention is required. However, a combination of features in a search, would indicate a slow attentional process. Thus, this discrepancy illustrates how speed and accuracy directly link to the intensity of the attentional resources being engaged.

Method

Participant

A total of 5 participants took part in the experiment. All the participants were female and between the ages of 19 to 20 and a total of 200 trials were conducted for each participant. All of them were acknowledged with the purpose of the experiment, i.e., to

see how individuals identify and locate specific targets within a visual display, understanding factors influencing attention, perception, and search efficiency. Consent was provided.

Materials & Procedure

The Visual Search experiment on a laptop, using the PsychoPy v2025.1.1. The procedure evaluated the participants' ability to determine and locate a target amongst distractors.

Here, the target was the letter 'T', amongst scattered letters "L" ; to help the participant focus a fixation cross was presented in the middle of the screen. Both the target and the distractors were in different positions for every trial; to avoid the predictability of the accurate answers (Figure 1). The participants should be able to identify the position of the target as quickly and as accurately as possible by clicking on them.

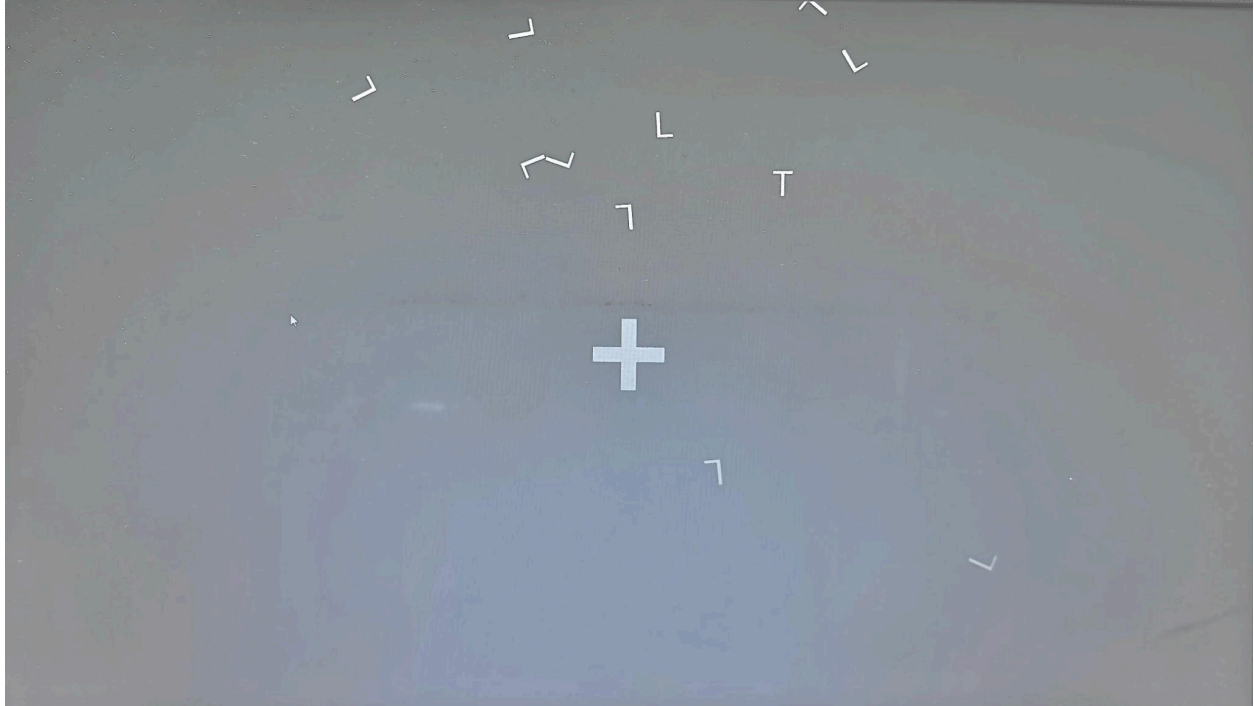
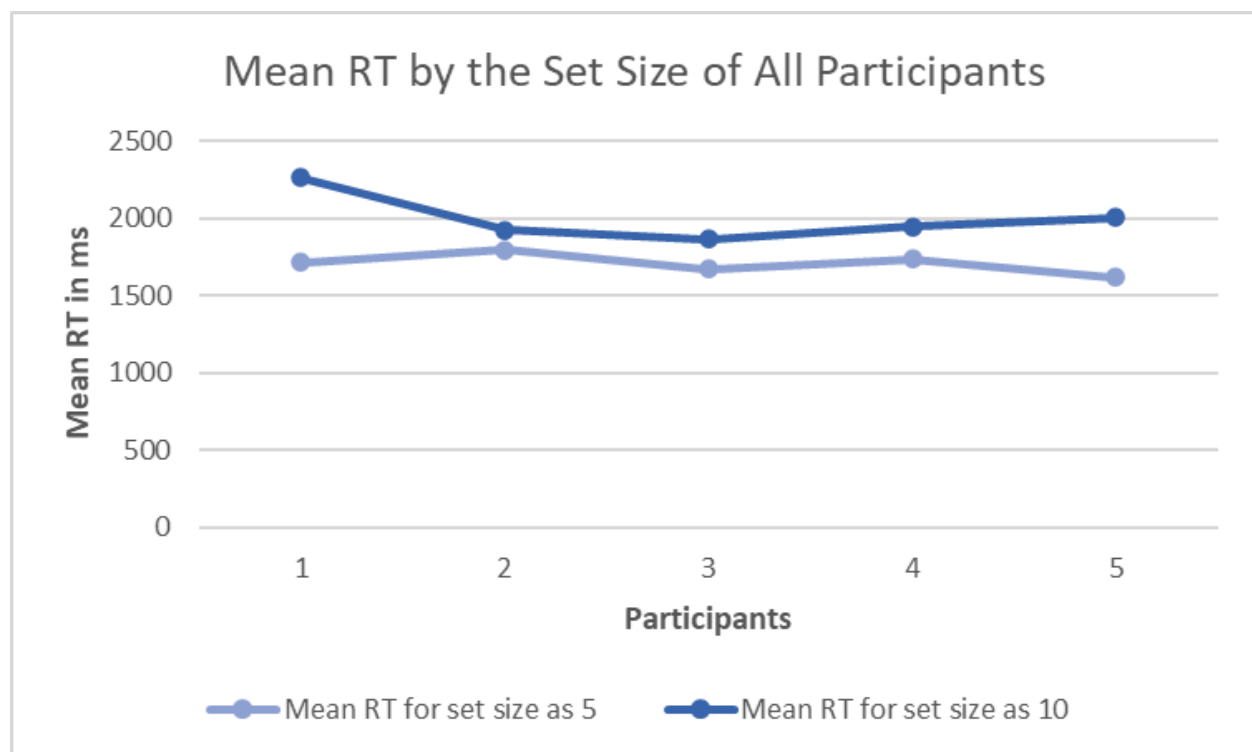


Figure 1 The target 'T' amongst the distractors and the fixation cross.

Results

The results show that there is a clear increase in Reaction Time (RT) as the number of distractors increases. This was consistent with all five participants for set size 10; that the mean reaction time was higher than for set size 5 - which confirms that the participants were not processing instantly, but rather taking time to process. The slope is a direct measure here of search efficiency and for us to find out what attention processes are underlying and being used. Across participants, the slopes indicate

variability. For instance, Participant 1 slope (109.31 ms/item) suggests the individual was processing one item after the other whilst taking time (strong effect of distractors) and indicating a steeper slope. Whereas, Participants 2, 3 and 4 showed similar responses (Slopes: 24.88, 39.42, and 41.84 ms/item) which tells us that they have efficiently searched within the items, but a smaller effect of distractors. Participant 5's slope (77.58 ms/item); falling in the middle, indicates a moderately inefficient search. The differences in slopes between participants are normal and can be due to several factors, for example, individual differences or using of a strategy. This increase in RT with set size aligns with findings in the visual search paradigm, suggesting that search becomes more attentionally demanding as the number of items/distractors rises.



Discussions

The slope is a significant indicator of whether attention is step by step or all at once.

The data moves us from a simple description of behavior to a deeper analysis about processing and strategies. The results are consistent with the visual search paradigm, where reaction time increases as the number of items increases. The slope of RT and set size gives us the 'mental cost' of adding more items into the search. However, the data demonstrated the individual differences amongst the search efficiency. For example, Participant 1's search tells us that it was a strict search, based on the effect of the distractors. At the same time, some participants' data suggests that they might have adopted or formed strategies to search.

References

1. Kantowitz, B. H., Roediger, H. L., & Elmes, D. G. (2009). *Experimental psychology*. Boston, MA, USA: Wadsworth Cengage Learning.
2. Palmer, J., Ames, C. T., & Lindsey, D. T. (1993). Measuring the effect of attention on simple visual search. *Journal of Experimental Psychology: Human Perception and Performance*, 19(1), 108.